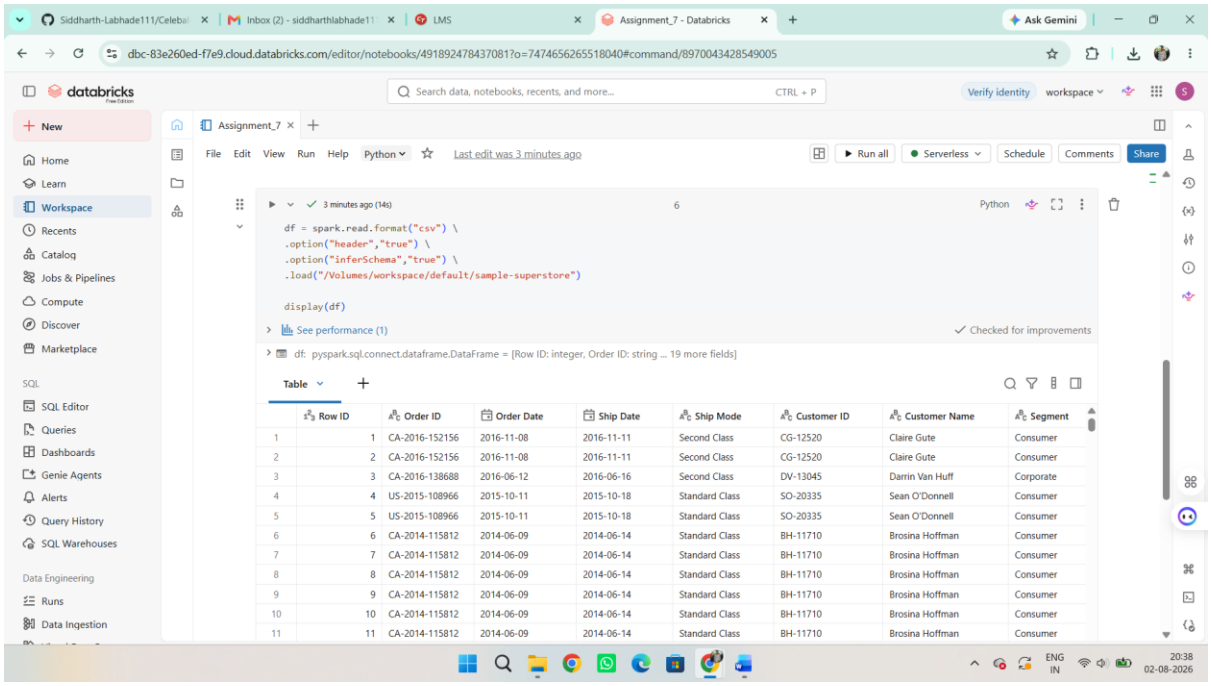
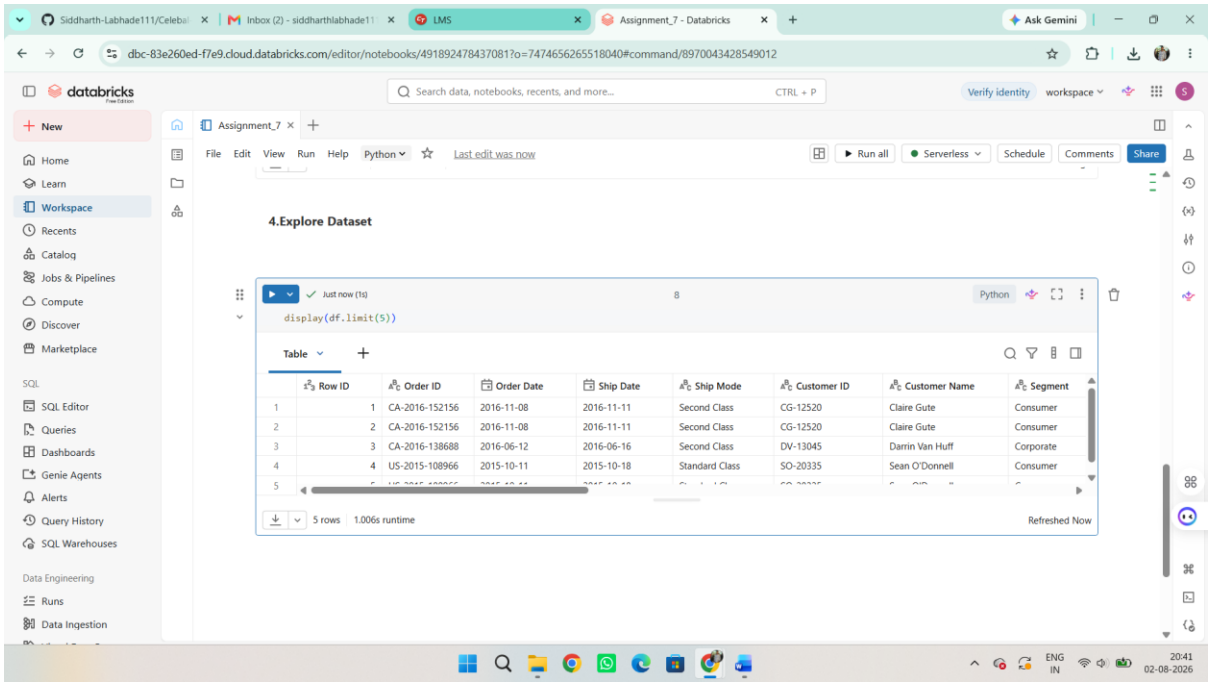
Week 7 Assignment

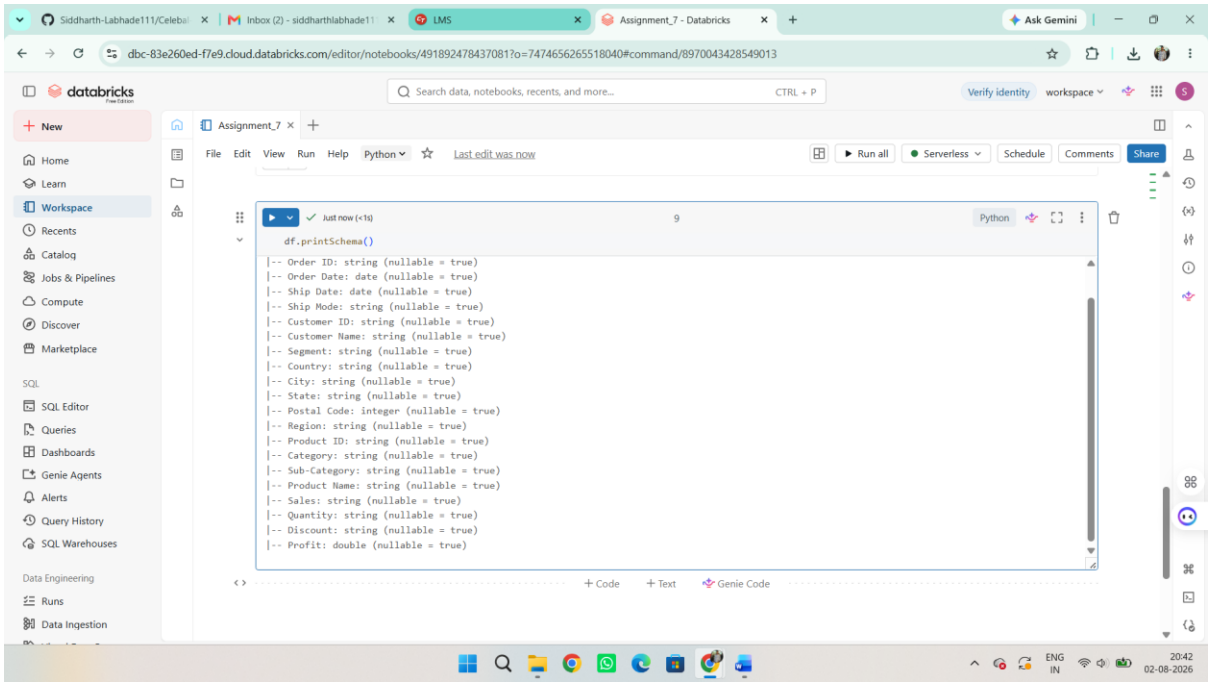
Name: Labhade Siddharth Popat
Student ID: CT_CSI_DE_1176
Dataset: Sample_Superstore

Assignment: Delta Lake MERGE Implementation

ScreenShots:




Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/8970043428549012
Assignment_7
File Edit View Run Help Python
Last edit was now
Run all
Serverless
Schedule
Comments
Share
4.Explore Dataset
display(df.limit(5))
Table
Row ID
Order ID
Order Date
Ship Date
Ship Mode
Customer ID
Customer Name
Segment
1 CA-2016-152156 2016-11-08 2016-11-11 Second Class CG-12520 Claire Gute Consumer
2 CA-2016-152156 2016-11-08 2016-11-11 Second Class CG-12520 Claire Gute Consumer
3 CA-2016-138688 2016-06-12 2016-06-16 Second Class DV-13045 Darrin Van Huff Corporate
4 US-2015-108966 2015-10-11 2015-10-18 Standard Class SO-20335 Sean O'Donnell Consumer
5 rows 1.006s runtime
Refreshed Now


Assignment_7 - Databricks
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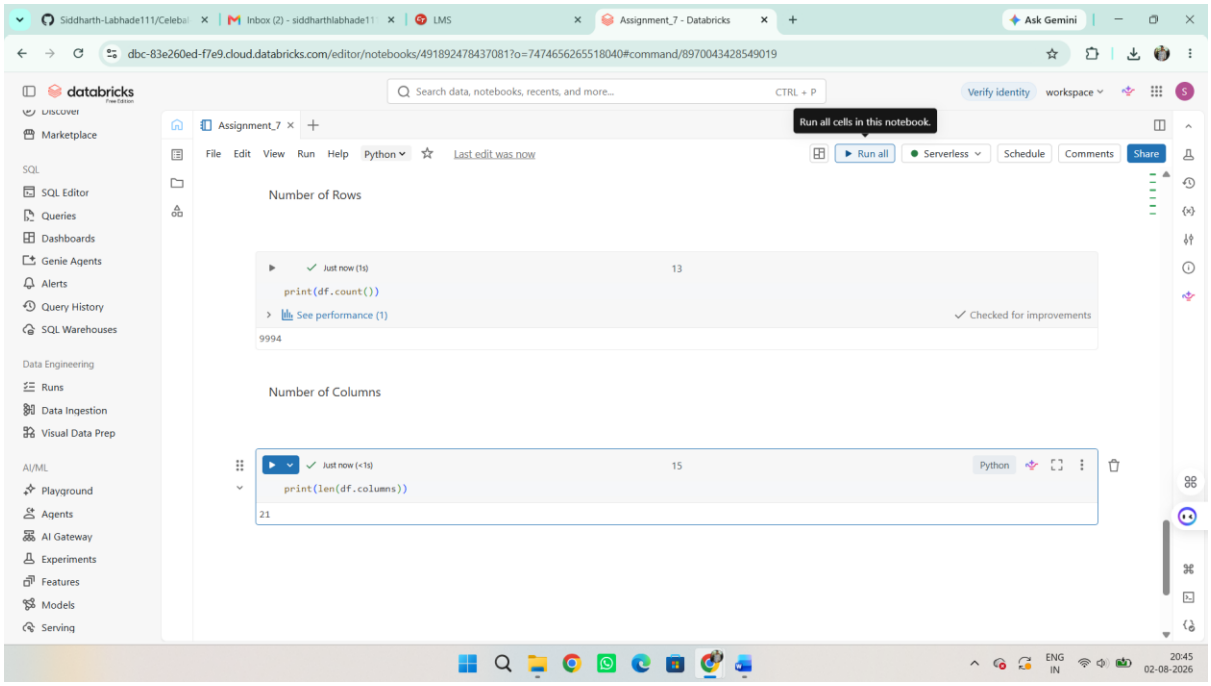
Assignment_7
File Edit View Run Help Python
Last edit was now
Run all
Serverless
Schedule
Comments
Share
df.printSchema()
|-- Order ID: string (nullable = true)
|-- Order Date: date (nullable = true)
|-- Ship Date: date (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal Code: integer (nullable = true)
|-- Region: string (nullable = true)
|-- Product ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product Name: string (nullable = true)
|-- Sales: string (nullable = true)
|-- Quantity: string (nullable = true)
|-- Discount: string (nullable = true)
|-- Profit: double (nullable = true)
+ Code
+ Text
Genie Code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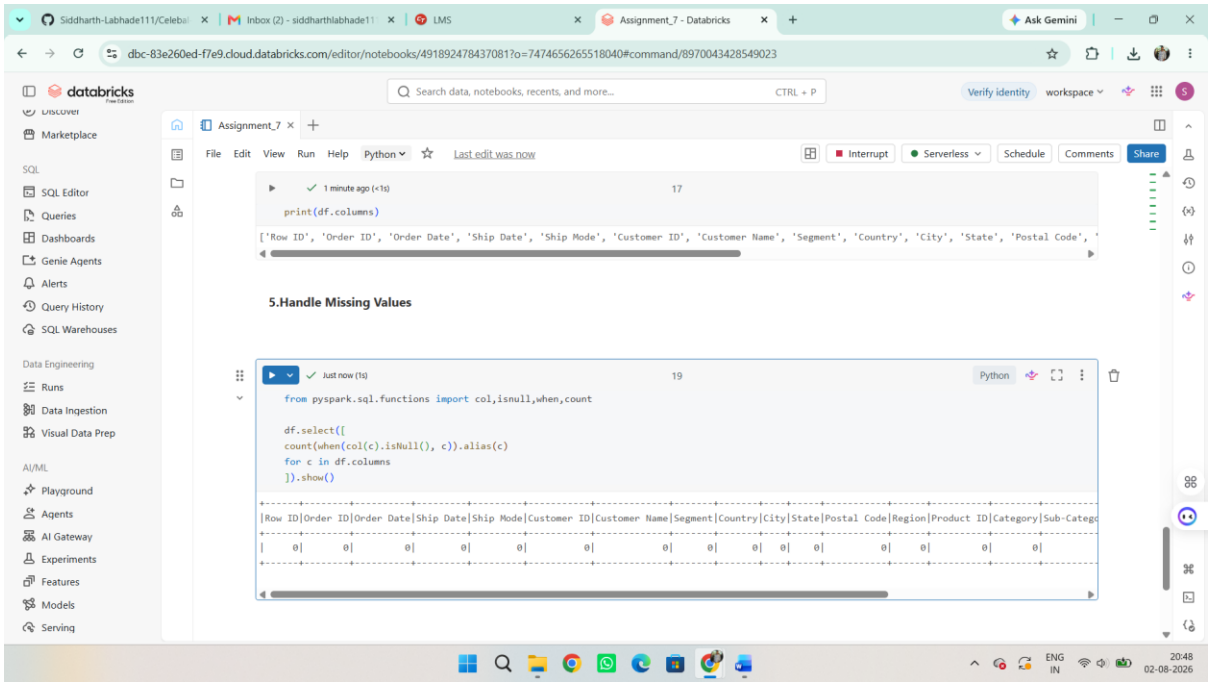
Run all cells in this notebook.
Assignment_7
File Edit View Run Help Python
Last edit was now
Run all
Serverless
Schedule
Comments
Share
Number of Rows
Just now (1s)
13
print(df.count())
See performance (1)
Checked for improvements
9994
Number of Columns
Just now (<1s)
15
Python
print(len(df.columns))
21


Assignment_7 - Databricks
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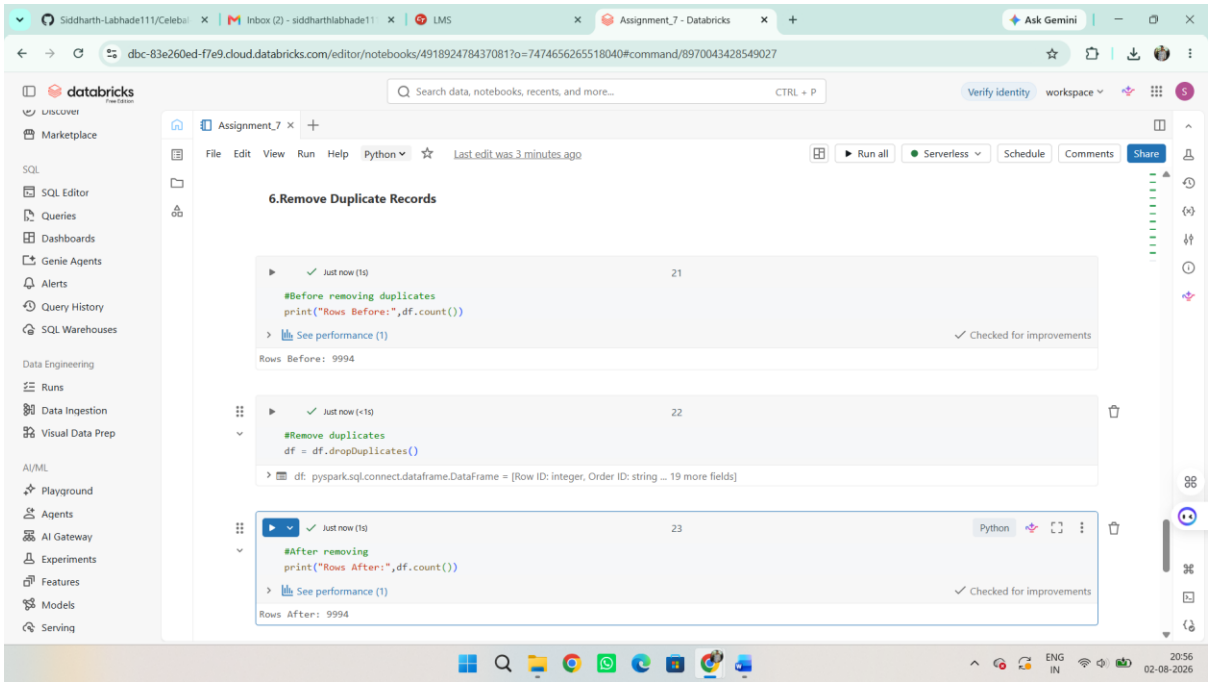
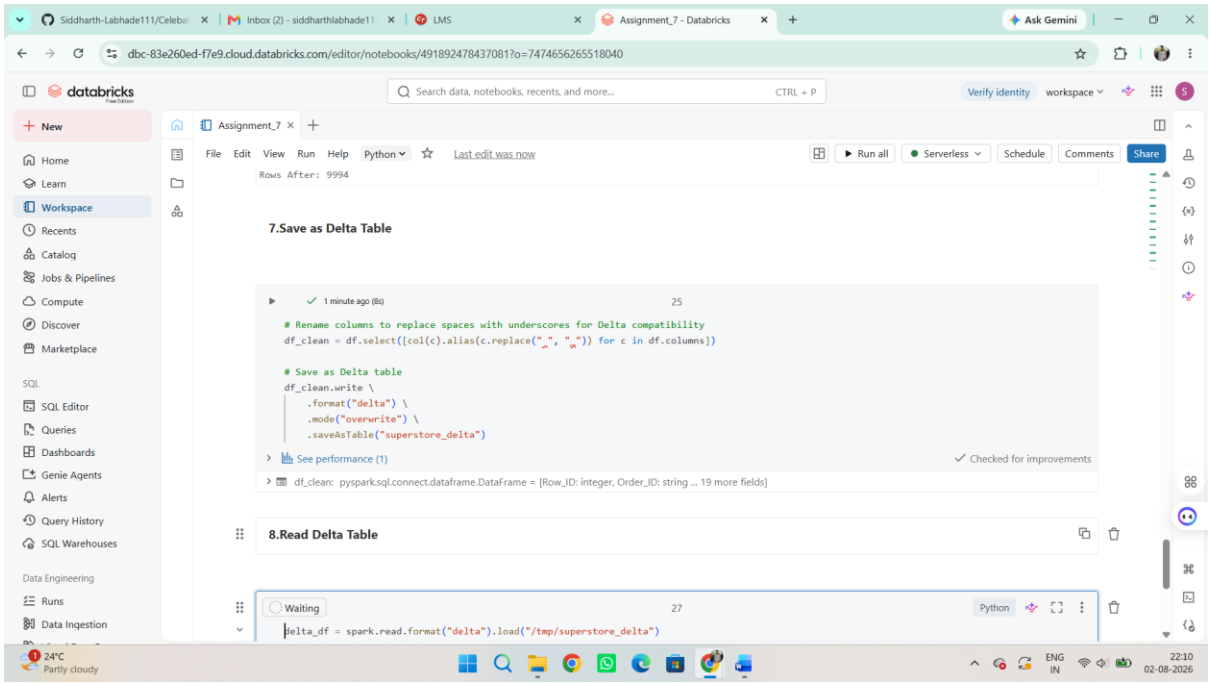
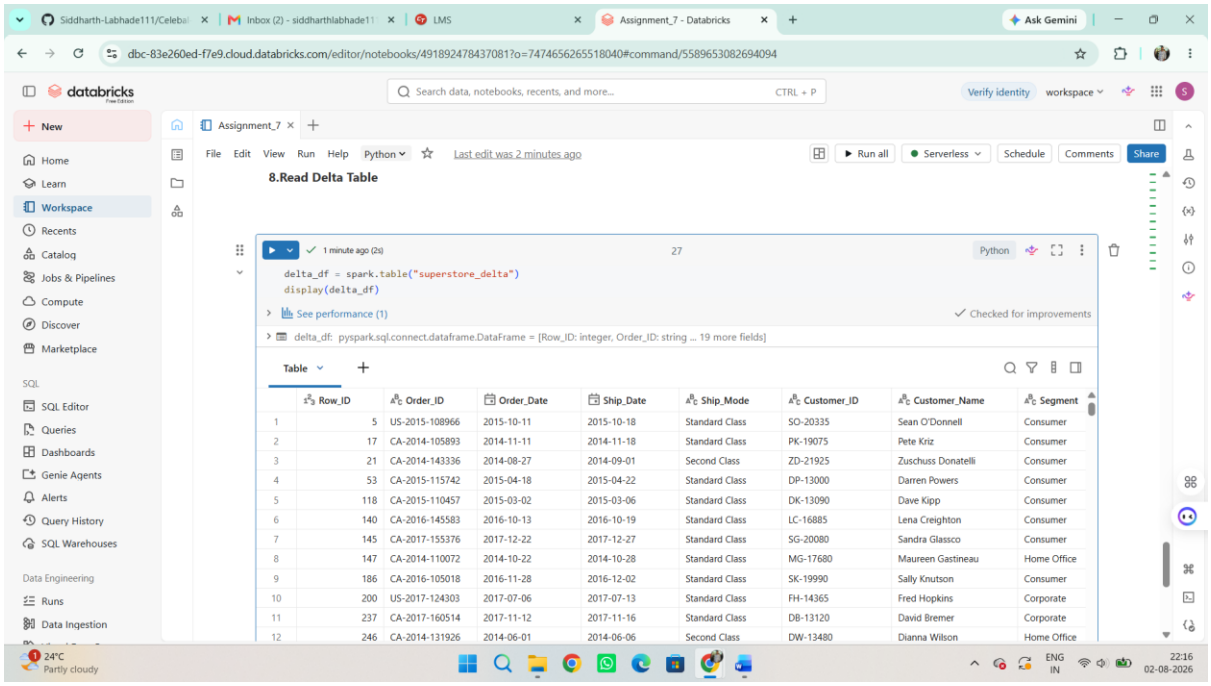
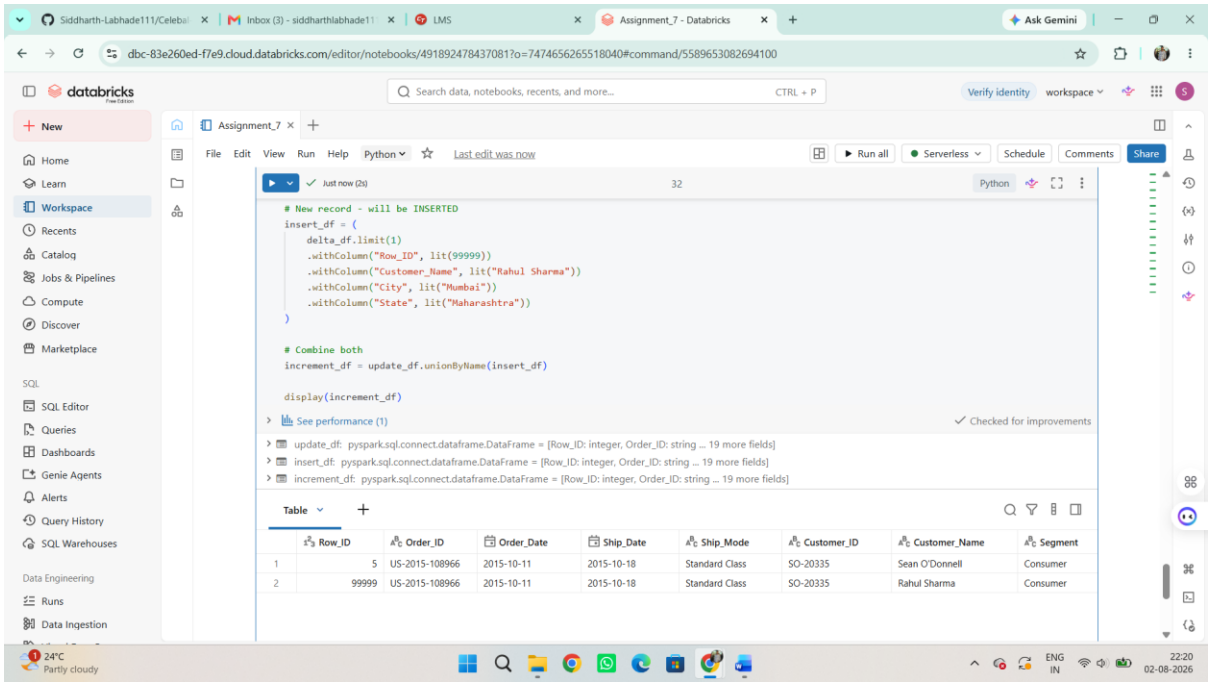
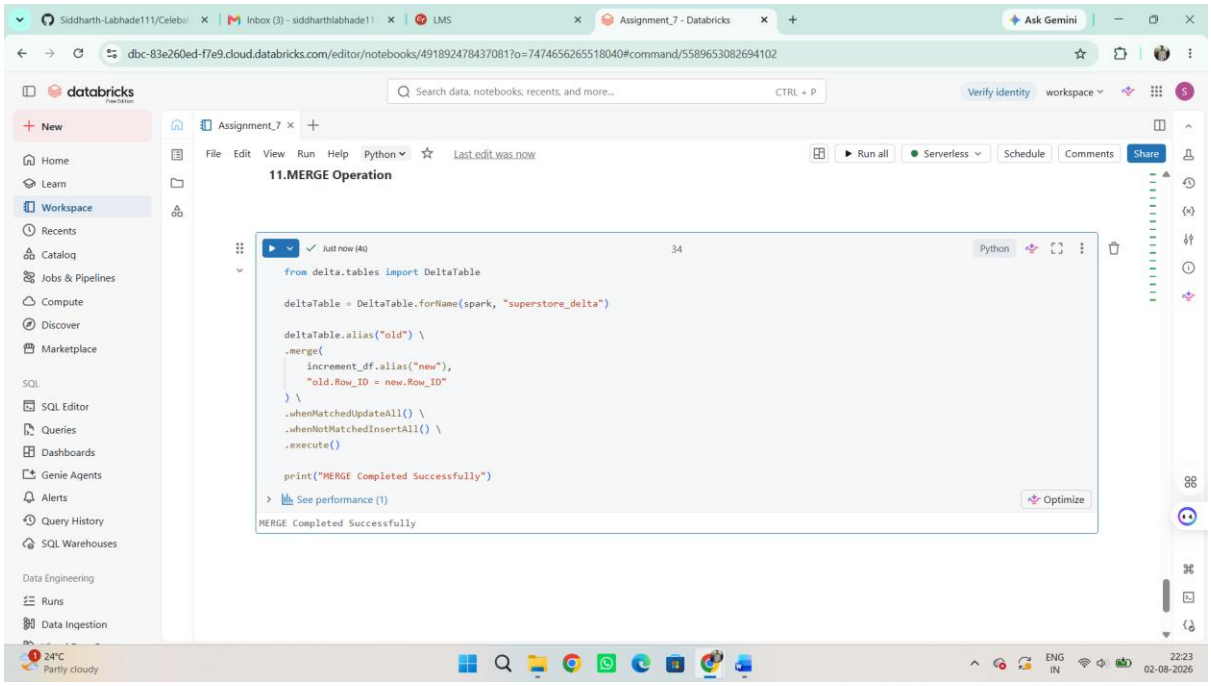
Assignment_7
File Edit View Run Help Python
Last edit was now
Interrupt
Serverless
Schedule
Comments
Share
1 minute ago (<1s)
17
print(df.columns)
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code',
5.Handle Missing Values
Just now (1s)
19
Python
from pyspark.sql.functions import col,isnull,when,count
df.select([
count(when(col(c).isNull(), c)).alias(c)
for c in df.columns
]).show()
|Row ID|Order ID|Order Date|Ship Date|Ship Mode|Customer ID|Customer Name|Segment|Country|City|State|Postal Code|Region|Product ID|Category|Sub-Categ
| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0|

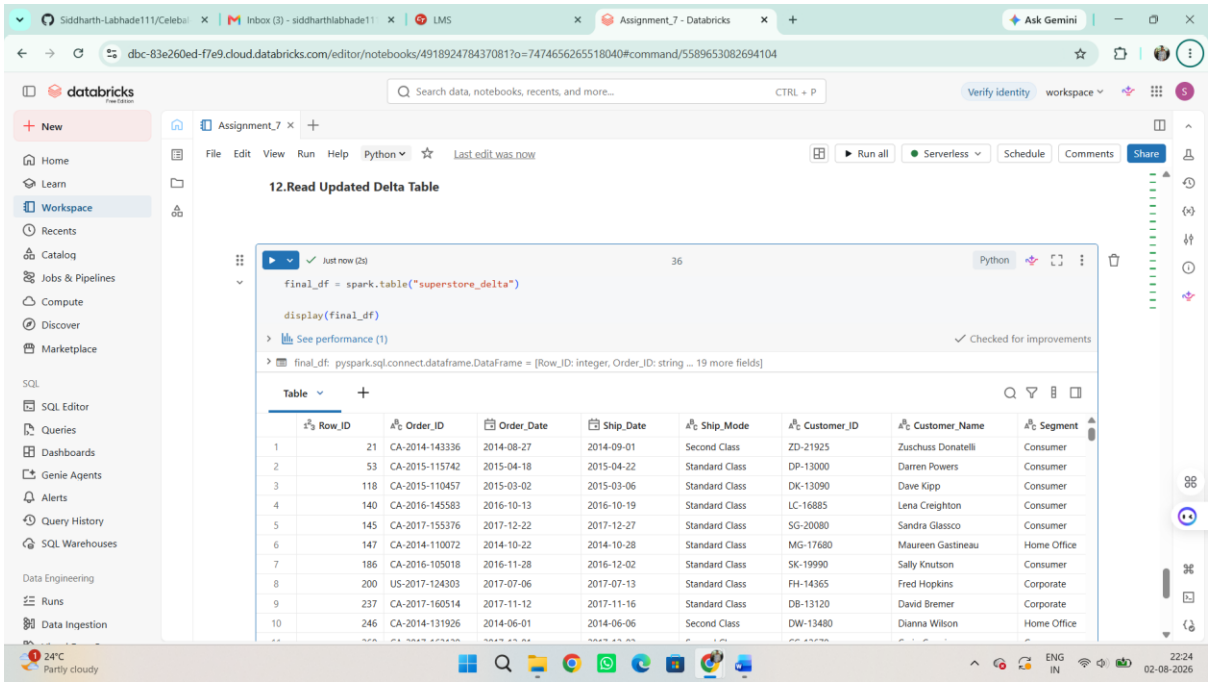

Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/8970043428549027
Assignment_7
File Edit View Run Help Python
Last edit was 3 minutes ago
Run all
Serverless
Schedule
Comments
Share
6.Remove Duplicate Records
#Before removing duplicates
print("Rows Before:",df.count())
Rows Before: 9994
#Remove duplicates
df = df.dropDuplicates()
#After removing
print("Rows After:",df.count())
Rows After: 9994

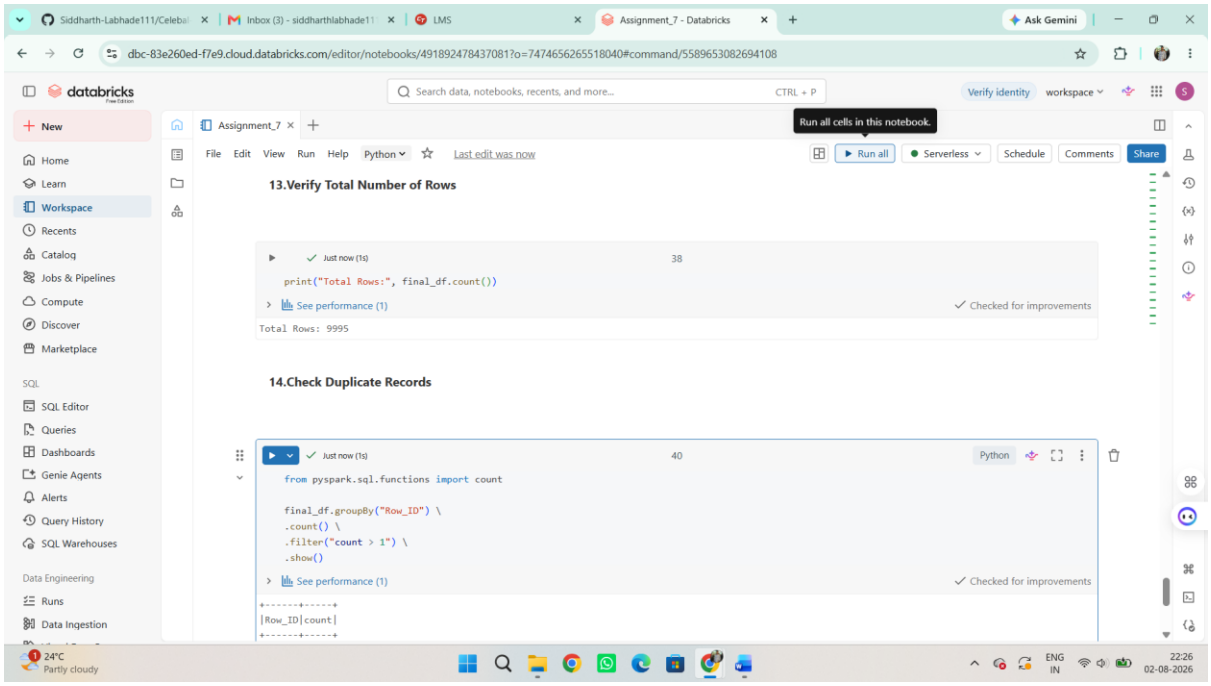

Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040
Assignment_7
File Edit View Run Help Python
Last edit was now
Rows After: 9994
7.Save as Delta Table
# Rename columns to replace spaces with underscores for Delta compatibility
df_clean = df.select([col(c).alias(c.replace(" ", "_")) for c in df.columns])
# Save as Delta table
df_clean.write \
.format("delta") \
.mode("overwrite") \
.saveAsTable("superstore_delta")
8.Read Delta Table
delta_df = spark.read.format("delta").load("/tmp/superstore_delta")


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694094
Assignment_7
File Edit View Run Help Python
Last edit was 2 minutes ago
Run all
Serverless
Schedule
Comments
Share
8.Read Delta Table
1 minute ago (2s)
27
Python
delta_df = spark.table("superstore_delta")
display(delta_df)
See performance (1)
Checked for improvements
delta_df: pyspark.sql.connect.dataframe.DataFrame = [Row_ID: integer, Order_ID: string ... 19 more fields]
Table
Row_ID
Order_ID
Order_Date
Ship_Date
Ship_Mode
Customer_ID
Customer_Name
Segment
1 5 US-2015-108966 2015-10-11 2015-10-18 Standard Class SO-20335 Sean O'Donnell Consumer
2 17 CA-2014-105893 2014-11-11 2014-11-18 Standard Class PK-19075 Pete Kriz Consumer
3 21 CA-2014-143336 2014-08-27 2014-09-01 Second Class ZD-21925 Zuschuss Donatelli Consumer
4 53 CA-2015-115742 2015-04-18 2015-04-22 Standard Class DP-13000 Darren Powers Consumer
5 118 CA-2015-110457 2015-03-02 2015-03-06 Standard Class DK-13090 Dave Kipp Consumer
6 140 CA-2016-145583 2016-10-13 2016-10-19 Standard Class LC-16885 Lena Creighton Consumer
7 145 CA-2017-155376 2017-12-22 2017-12-27 Standard Class SG-20080 Sandra Glassco Consumer
8 147 CA-2014-110072 2014-10-22 2014-10-28 Standard Class MG-17680 Maureen Gastineau Home Office
9 186 CA-2016-105018 2016-11-28 2016-12-02 Standard Class SK-19990 Sally Knutson Consumer
10 200 US-2017-124303 2017-07-06 2017-07-13 Standard Class FH-14365 Fred Hopkins Corporate
11 237 CA-2017-160514 2017-11-12 2017-11-16 Standard Class DB-13120 David Bremer Corporate
12 246 CA-2014-131926 2014-06-01 2014-06-06 Second Class DW-13480 Dianna Wilson Home Office


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694100
Assignment_7
File Edit View Run Help Python
Last edit was now
Run all
Serverless
Schedule
Comments
Share
Just now (2s)
32
Python
# New record - will be INSERTED
insert_df = (
delta_df.limit(1)
.withColumn("Row_ID", lit(99999))
.withColumn("Customer_Name", lit("Rahul Sharma"))
.withColumn("City", lit("Mumbai"))
.withColumn("State", lit("Maharashtra"))
)
# Combine both
increment_df = update_df.unionByName(insert_df)
display(increment_df)
See performance (1)
Checked for improvements
update_df: pyspark.sql.connect.dataframe.DataFrame = [Row_ID: integer, Order_ID: string ... 19 more fields]
insert_df: pyspark.sql.connect.dataframe.DataFrame = [Row_ID: integer, Order_ID: string ... 19 more fields]
increment_df: pyspark.sql.connect.dataframe.DataFrame = [Row_ID: integer, Order_ID: string ... 19 more fields]
Table
Row_ID
Order_ID
Order_Date
Ship_Date
Ship_Mode
Customer_ID
Customer_Name
Segment
1 5 US-2015-108966 2015-10-11 2015-10-18 Standard Class SO-20335 Sean O'Donnell Consumer
2 99999 US-2015-108966 2015-10-11 2015-10-18 Standard Class SO-20335 Rahul Sharma Consumer


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694102
databricks
Assignment_7
11.MERGE Operation
Just now (4s)
34
Python
from delta.tables import DeltaTable
deltaTable = DeltaTable.forName(spark, "superstore_delta")
deltaTable.alias("old") \
.merge(
increment_df.alias("new"),
"old.Row_ID = new.Row_ID"
) \
.whenMatchedUpdateAll() \
.whenNotMatchedInsertAll() \
.execute()
print("MERGE Completed Successfully")
See performance (1)
Optimize
MERGE Completed Successfully


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694104
databricks
Assignment_7
12.Read Updated Delta Table
Just now (2s)
36
Python
final_df = spark.table("superstore_delta")
display(final_df)
See performance (1)
Checked for improvements
final_df: pyspark.sql.connect.dataframe.DataFrame = [Row_ID: integer, Order_ID: string ... 19 more fields]
Table
Row_ID
Order_ID
Order_Date
Ship_Date
Ship_Mode
Customer_ID
Customer_Name
Segment
1 21 CA-2014-143336 2014-08-27 2014-09-01 Second Class ZD-21925 Zuschuss Donatelli Consumer
2 53 CA-2015-115742 2015-04-18 2015-04-22 Standard Class DP-13000 Darren Powers Consumer
3 118 CA-2015-110457 2015-03-02 2015-03-06 Standard Class DK-13090 Dave Kipp Consumer
4 140 CA-2016-145583 2016-10-13 2016-10-19 Standard Class LC-16885 Lena Creighton Consumer
5 145 CA-2017-155376 2017-12-22 2017-12-27 Standard Class SG-20080 Sandra Glassco Consumer
6 147 CA-2014-110072 2014-10-22 2014-10-28 Standard Class MG-17680 Maureen Gastineau Home Office
7 186 CA-2016-105018 2016-11-28 2016-12-02 Standard Class SK-19990 Sally Knutson Consumer
8 200 US-2017-124303 2017-07-06 2017-07-13 Standard Class FH-14365 Fred Hopkins Corporate
9 237 CA-2017-160514 2017-11-12 2017-11-16 Standard Class DB-13120 David Bremer Corporate
10 246 CA-2014-131926 2014-06-01 2014-06-06 Second Class DW-13480 Dianna Wilson Home Office


Assignment_7 - Databricks
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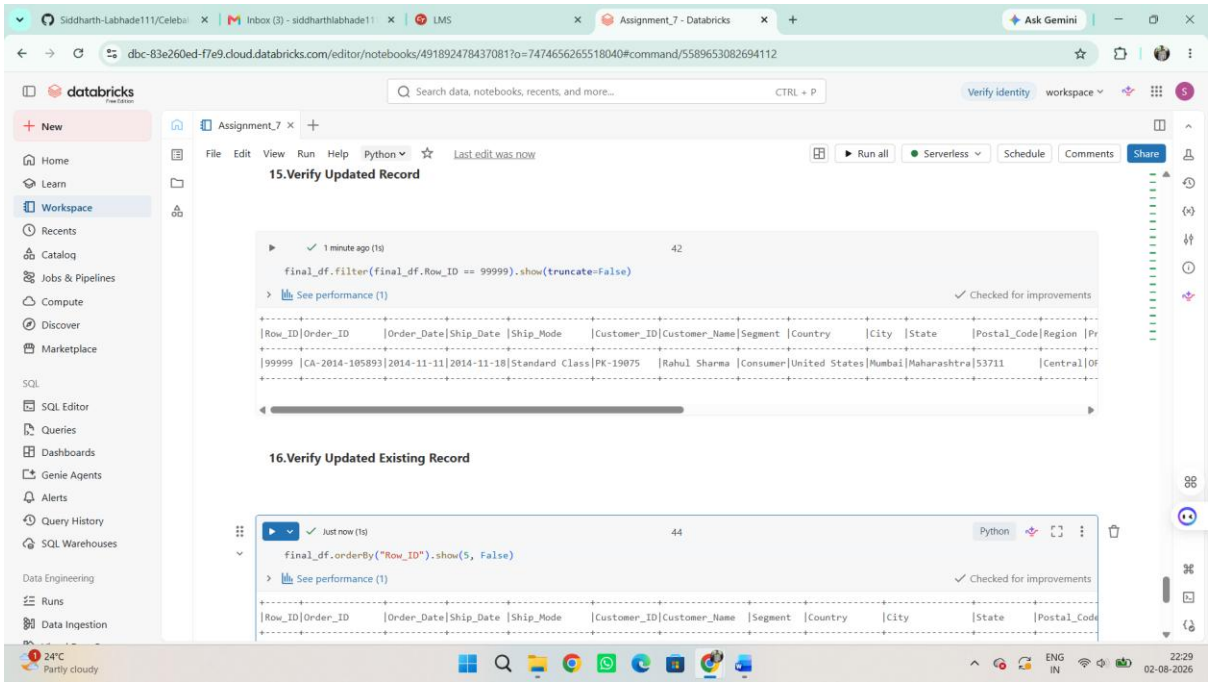
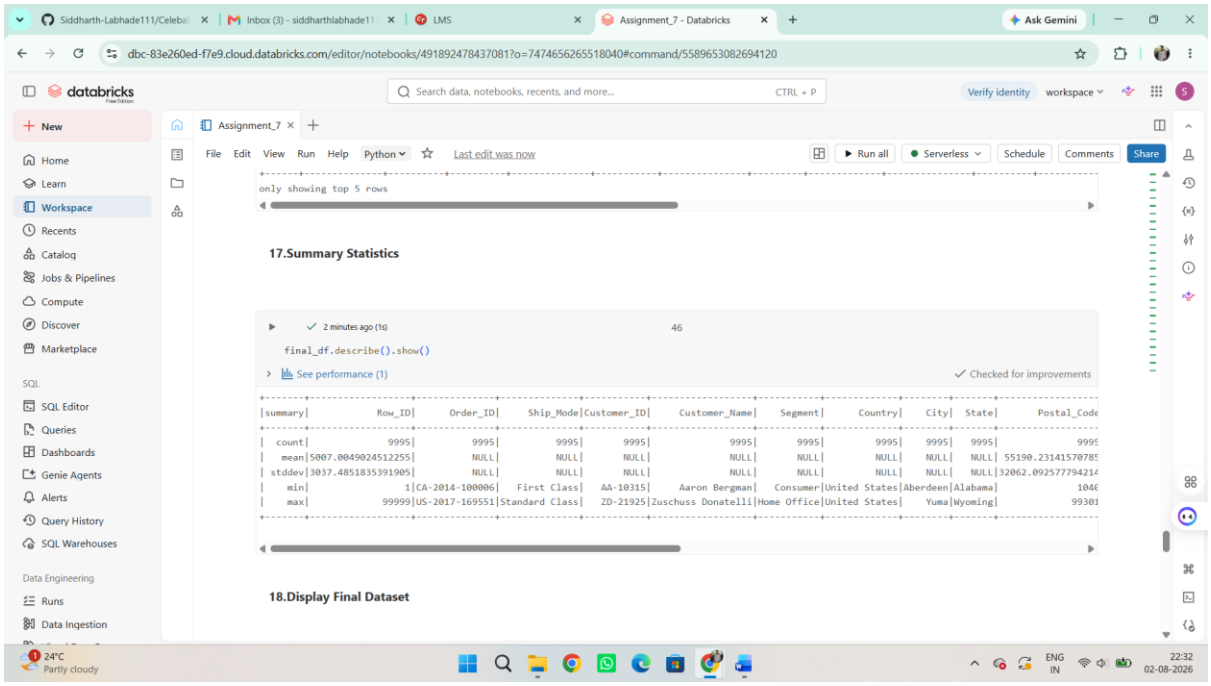
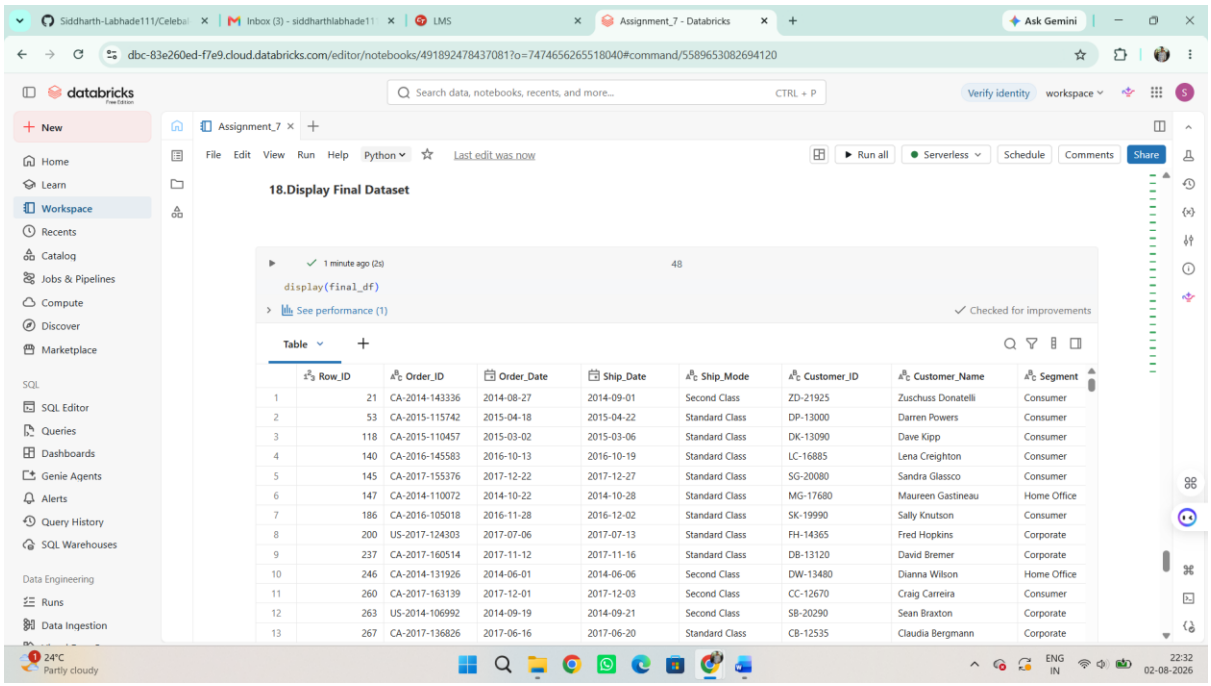
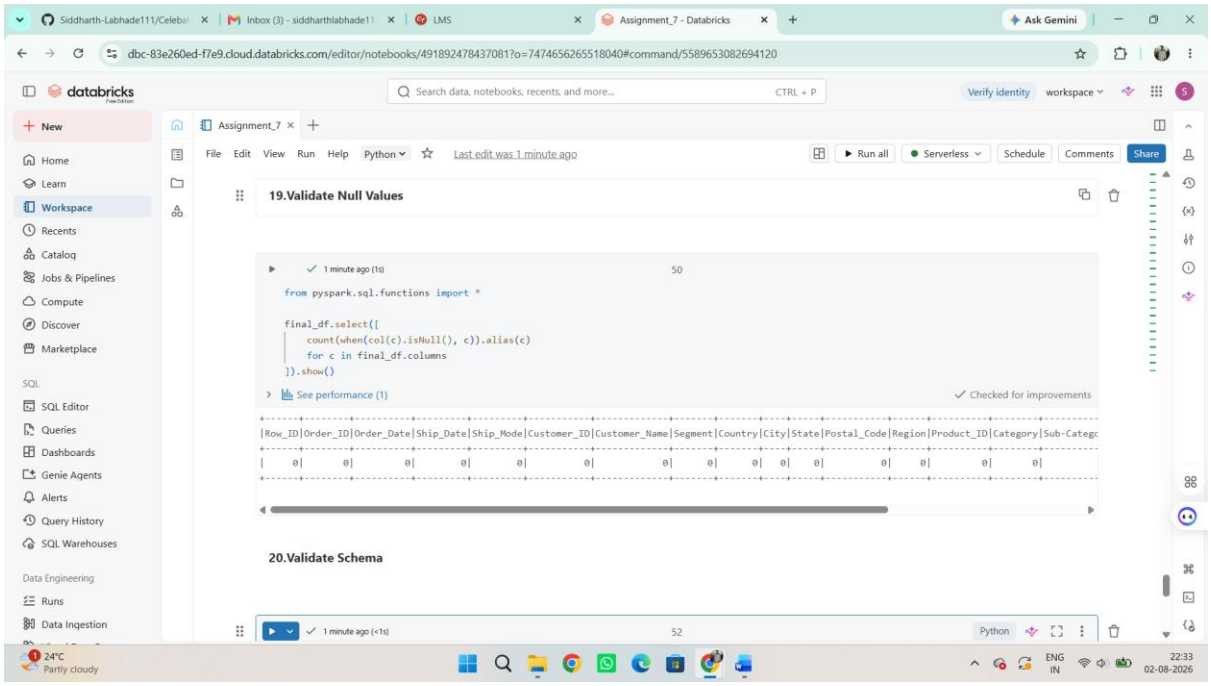
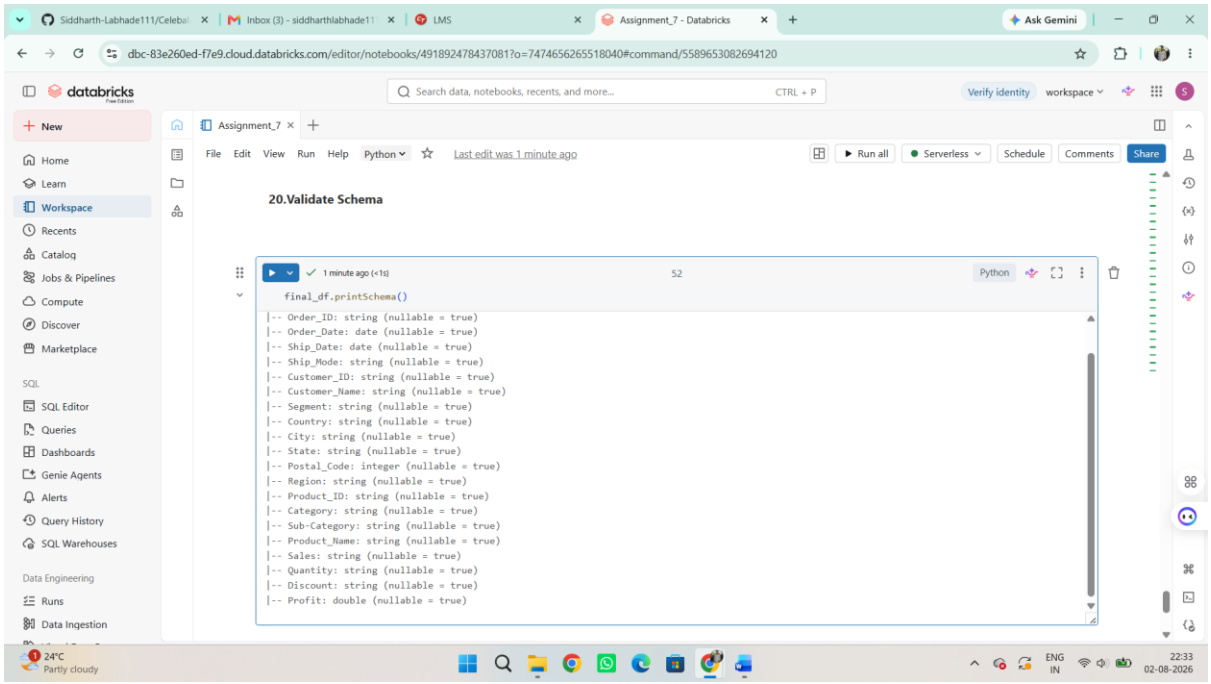
Assignment_7
Run all cells in this notebook.
13.Verify Total Number of Rows
print("Total Rows:", final_df.count())
Total Rows: 9995
14.Check Duplicate Records
from pyspark.sql.functions import count
final_df.groupBy("Row_ID") \
.count() \
.filter("count > 1") \
.show()
|Row_ID|count|


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694112
Assignment_7
15.Verify Updated Record
final_df.filter(final_df.Row_ID == 99999).show(truncate=False)
|Row_ID|Order_ID |Order_Date|Ship_Date |Ship_Mode |Customer_ID|Customer_Name|Segment |Country |City |State |Postal_Code|Region |
|99999 |CA-2014-105893|2014-11-11|2014-11-18|Standard Class|PK-19075 |Rahul Sharma |Consumer|United States|Mumbai|Maharashtra|53711 |Central|
16.Verify Updated Existing Record
final_df.orderBy("Row_ID").show(5, False)
|Row_ID|Order_ID |Order_Date|Ship_Date |Ship_Mode |Customer_ID|Customer_Name |Segment |Country |City |State |Postal_Code


only showing top 5 rows
17.Summary Statistics
2 minutes ago (1s)
46
final_df.describe().show()
See performance (1)
Checked for improvements
|summary| Row_ID| Order_ID| Ship_Mode|Customer_ID| Customer_Name| Segment| Country| City| State| Postal_Code
| count| 9995| 9995| 9995| 9995| 9995| 9995| 9995| 9995| 9995| 9995
| mean|5007.0049024512255| NULL| NULL| NULL| NULL| NULL| NULL| NULL| NULL| 55190.23141570785
| stddev|3037.4851835391905| NULL| NULL| NULL| NULL| NULL| NULL| NULL| NULL|32062.092577794214
| min| 1|CA-2014-100006| First Class| AA-10315| Aaron Bergman| Consumer|United States|Aberdeen|Alabama| 1040
| max| 99999|US-2017-169551|Standard Class| ZD-21925|Zuschuss Donatelli|Home Office|United States| Yuma|Wyoming| 99301
18.Display Final Dataset


18.Display Final Dataset
1 minute ago (2s)
48
display(final_df)
See performance (1)
Checked for improvements
Table
Row_ID Order_ID Order_Date Ship_Date Ship_Mode Customer_ID Customer_Name Segment
1 21 CA-2014-143336 2014-08-27 2014-09-01 Second Class ZD-21925 Zuschuss Donatelli Consumer
2 53 CA-2015-115742 2015-04-18 2015-04-22 Standard Class DP-13000 Darren Powers Consumer
3 118 CA-2015-110457 2015-03-02 2015-03-06 Standard Class DK-13090 Dave Kipp Consumer
4 140 CA-2016-145583 2016-10-13 2016-10-19 Standard Class LC-16885 Lena Creighton Consumer
5 145 CA-2017-155376 2017-12-22 2017-12-27 Standard Class SG-20080 Sandra Glassco Consumer
6 147 CA-2014-110072 2014-10-22 2014-10-28 Standard Class MG-17680 Maureen Gastineau Home Office
7 186 CA-2016-105018 2016-11-28 2016-12-02 Standard Class SK-19990 Sally Knutson Consumer
8 200 US-2017-124303 2017-07-06 2017-07-13 Standard Class FH-14365 Fred Hopkins Corporate
9 237 CA-2017-160514 2017-11-12 2017-11-16 Standard Class DB-13120 David Bremer Corporate
10 246 CA-2014-131926 2014-06-01 2014-06-06 Second Class DW-13480 Dianna Wilson Home Office
11 260 CA-2017-163139 2017-12-01 2017-12-03 Second Class CC-12670 Craig Carreira Consumer
12 263 US-2014-106992 2014-09-19 2014-09-21 Second Class SB-20290 Sean Braxton Corporate
13 267 CA-2017-136826 2017-06-16 2017-06-20 Standard Class CB-12535 Claudia Bergmann Corporate


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694120
databricks
Search data, notebooks, recents, and more... CTRL + P
Verify identity
workspace
New
Assignment_7
File Edit View Run Help Python
Last edit was 1 minute ago
Run all
Serverless
Schedule
Comments
Share
Home
Learn
Workspace
Recents
Catalog
Jobs & Pipelines
Compute
Discover
Marketplace
SQL
SQL Editor
Queries
Dashboards
Genie Agents
Alerts
Query History
SQL Warehouses
Data Engineering
Runs
Data Ingestion
19.Validate Null Values
1 minute ago (1s)
50
from pyspark.sql.functions import *
final_df.select([
count(when(col(c).isNull(), c)).alias(c)
for c in final_df.columns
]).show()
See performance (1)
Checked for improvements
|Row_ID|Order_ID|Order_Date|Ship_Date|Ship_Mode|Customer_ID|Customer_Name|Segment|Country|City|State|Postal_Code|Region|Product_ID|Category|Sub-Categc
| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0| 0|
20.Validate Schema
1 minute ago (<1s)
52
Python
24°C
Partly cloudy
ENG
IN
22:33
02-08-2026


Assignment_7 - Databricks
dbc-83e260ed-f7e9.cloud.databricks.com/editor/notebooks/4918924784370 81?o=7474656265518040#command/5589653082694120
databricks
Search data, notebooks, recents, and more... CTRL + P
Verify identity
workspace
New
Assignment_7
File Edit View Run Help Python
Last edit was 1 minute ago
Run all
Serverless
Schedule
Comments
Share
Home
Learn
Workspace
Recents
Catalog
Jobs & Pipelines
Compute
Discover
Marketplace
SQL
SQL Editor
Queries
Dashboards
Genie Agents
Alerts
Query History
SQL Warehouses
Data Engineering
Runs
Data Ingestion
20.Validate Schema
1 minute ago (<1s)
52
Python
final_df.printSchema()
|-- Order_ID: string (nullable = true)
|-- Order_Date: date (nullable = true)
|-- Ship_Date: date (nullable = true)
|-- Ship_Mode: string (nullable = true)
|-- Customer_ID: string (nullable = true)
|-- Customer_Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal_Code: integer (nullable = true)
|-- Region: string (nullable = true)
|-- Product_ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product_Name: string (nullable = true)
|-- Sales: string (nullable = true)
|-- Quantity: string (nullable = true)
|-- Discount: string (nullable = true)
|-- Profit: double (nullable = true)
24°C
Partly cloudy
ENG
IN
22:33
02-08-2026